Military Stress Analysis

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Project Overview

Stress is a critical psychological and physiological challenge across society. Prolonged stress can cause anxiety, depression, cardiovascular disease, burnout, sleep disturbance, and impaired cognition–risks amplified in high-risk professions like soldiers and first responders.

Why Multimodal Screening?

Single-modality Limits

Facial cues can be masked; voice analysis is sensitive to noise and language–both alone yield inconsistent results.

Multimodal Advantage

Combining facial and acoustic signals improves reliability, robustness, and accuracy for stress screening.

Project Objectives

Real-time Capture

Collect audio/video via standard webcam and microphone.

Feature Extraction

Extract facial (action units, gaze, micro-expressions) and speech (pitch, MFCC, prosody) features.

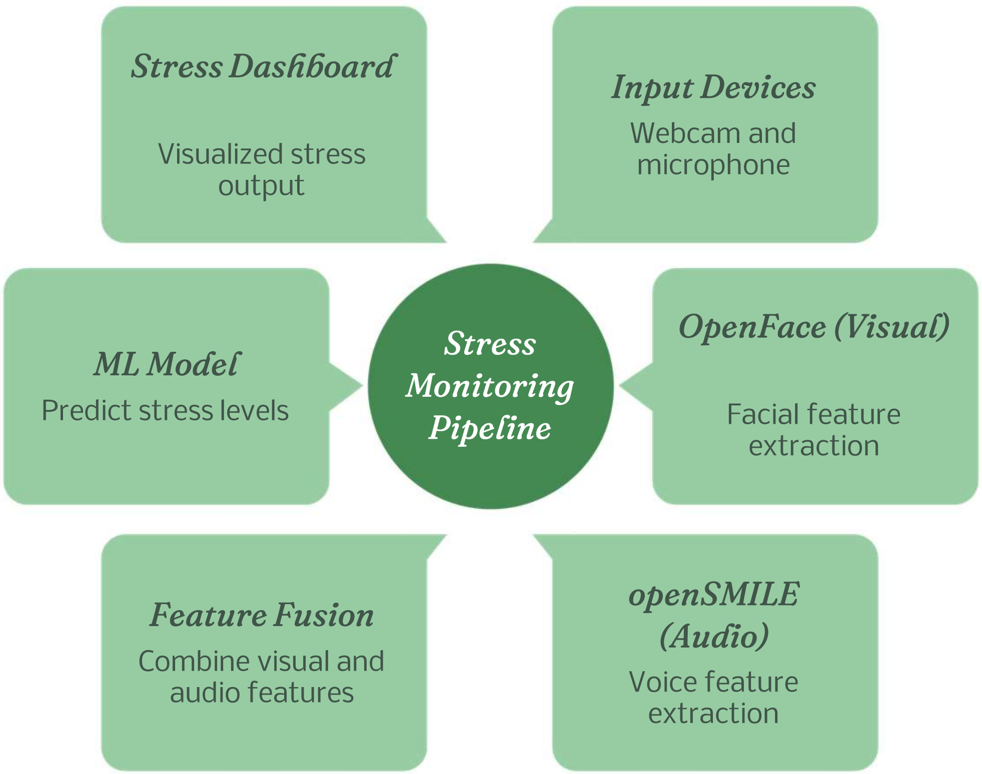
Fusion & Classification

Fuse modalities and train ML models (Random Forest, SVM, or Neural Network) to classify Low / Moderate / High stress.

Dashboard

Deliver a user-friendly dashboard showing predicted stress level and confidence score.

System Architecture



The pipeline: Input → OpenFace (visual) & openSMILE (audio) → Feature Fusion → ML Model → Stress Dashboard.

Feature Extraction

- Facial: action units, head pose, eye gaze, blink rate, micro-expressions
- Voice: pitch, intensity, MFCCs, prosody, speech rate

Methodology & Evaluation

Data

DAIC-WOZ corpus for synchronized audio, video, and labels.

Pipeline Steps

1) Data collection 2) Feature extraction 3) Preprocessing 4) Model training 5) Real-time prototype 6) Evaluation

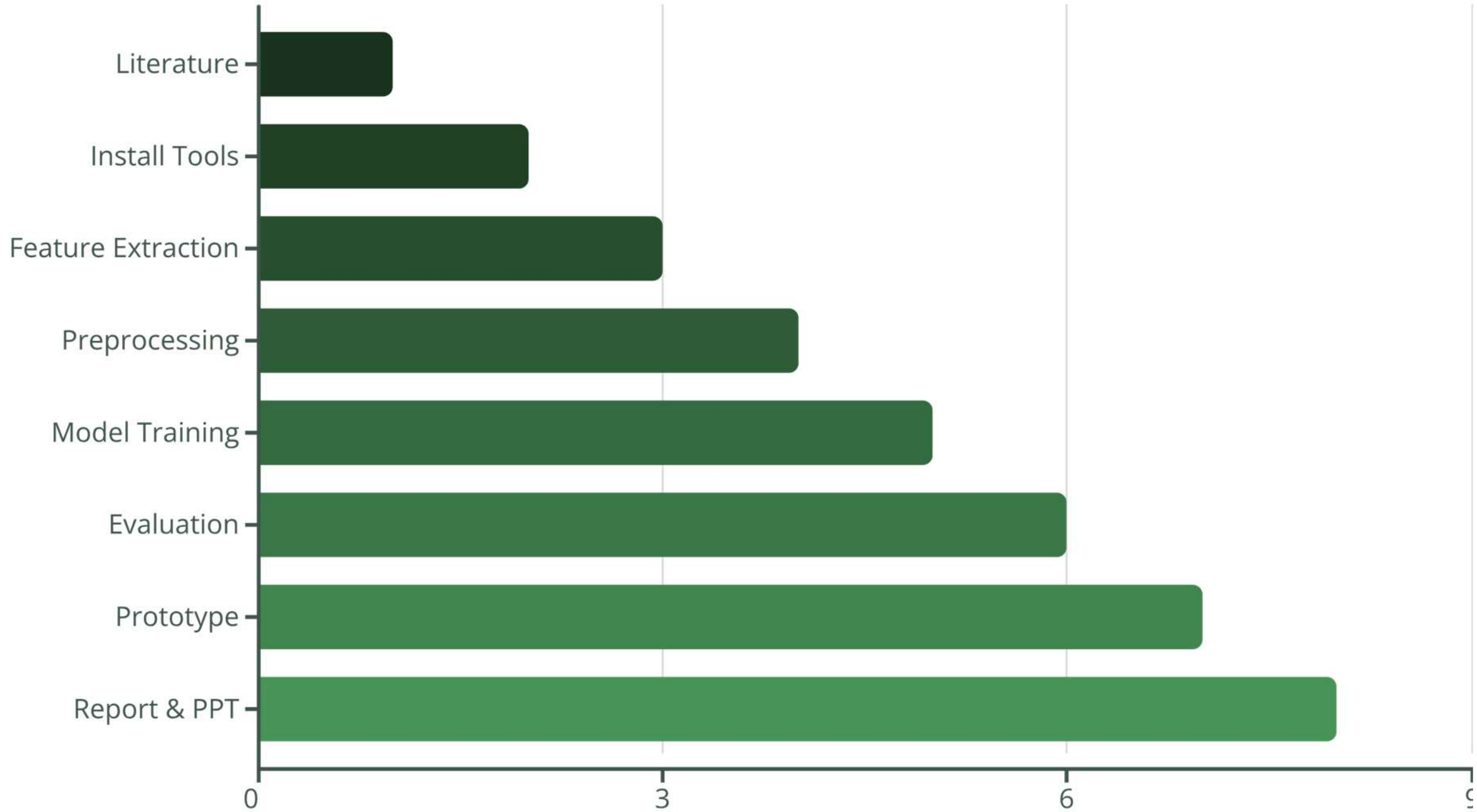
Metrics

Accuracy, precision, recall, F1-score, and confusion matrix.

Real-Time Prototype & Dashboard

The prototype captures live streams, extracts features, runs the trained model, and displays stress levels (Low / Moderate / High) with confidence–designed for nontechnical users in education, corporate, healthcare, and defense settings.

Project Timeline (8 Weeks)



The 8-week plan covers tool setup, extraction, modeling, prototyping, and final reporting.

Outcomes, Scope & Ethics

- Outcome: Real-time multimodal stress screening prototype and interactive dashboard.
- Scope: Low-cost, scalable–schools, workplaces, healthcare, defense; research tool for affective computing.
- Ethics: Screening only–not a medical diagnosis. No data stored without explicit consent; privacy, encryption, and anonymity prioritized.
- Future: Wearables, deep learning, mobile/cloud deployment.